



DATA-DRIVEN INSIGHTS

Customer Shopping Behavior Analysis

Identifying Patterns,
Driving Growth



Analysis



Segmentation



Growth



Project Overview



Records
3,900 Transactions



Demographics
Age, Gender, Location



Objective
Identify Spending Patterns

Project and Dataset Overview

Customer Demographics • Purchase Details • Shopping Behaviour



Customer
Demographics



Purchase
Details



Shopping
Behaviour



Feedback
Ratings

Key Dataset Metrics



Records

3,900

Transaction Rows



Features

18

Data Columns



Cleaned

37

Missing Values Fixed



Avg Rating

3.75

Out of 5.0

Exploratory Data Analysis

Standardization • Imputation • Engineering • Optimization



Standardization

Column names converted to snake_case



Missing Values

37 missing ratings filled with median



Feature Engineering

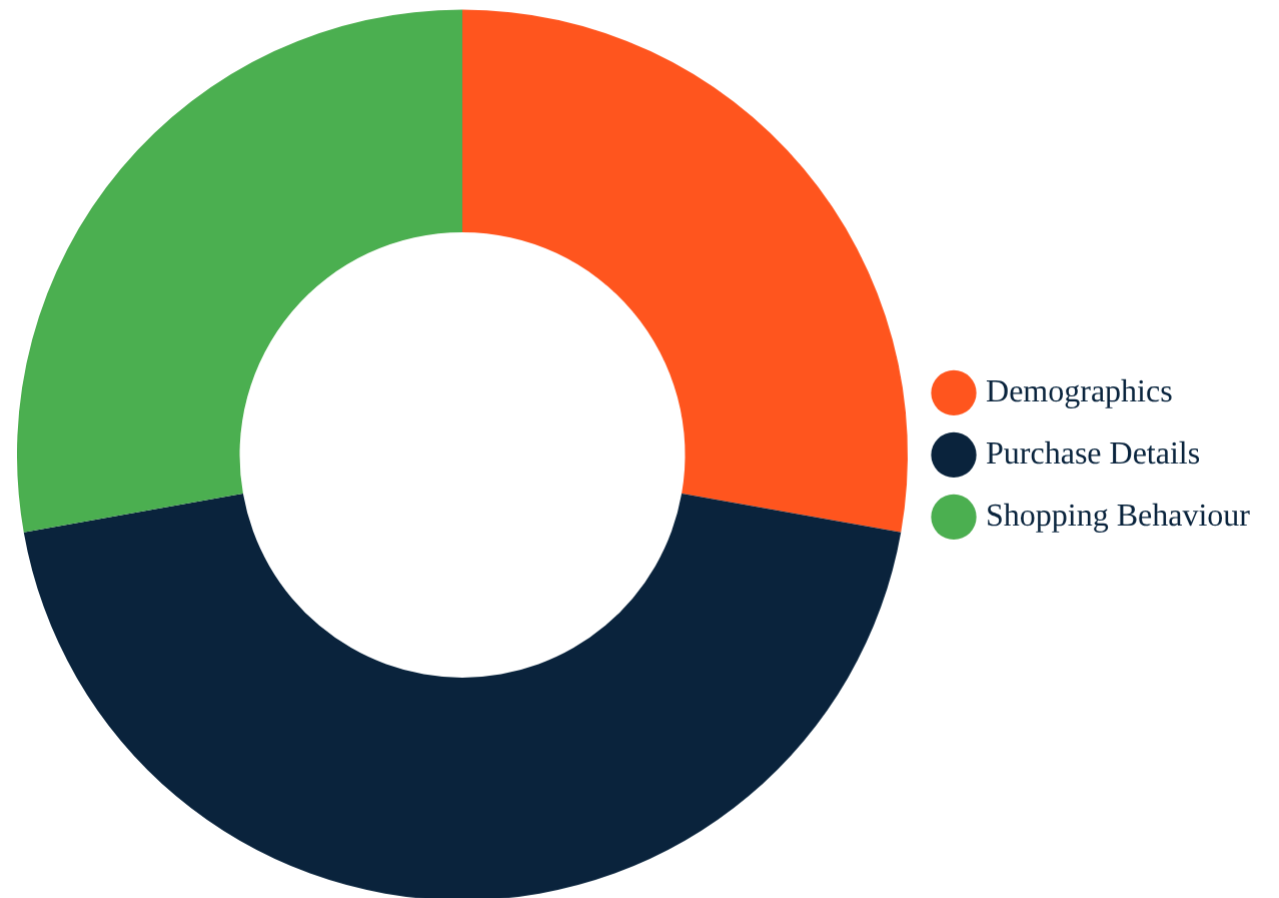
Added age_group & frequency columns



Redundancy Check

Dropped duplicate promo_code_used column

Dataset Column Distribution

**5**

Demographics

8

Purchase Details

5

Shopping Behavior

Core Personas

Demographics



Male Dominance

2x Revenue vs Female

Young Adults

Highest Revenue Segment



Location

Diverse Geographic Mix



High-Value Target

High Spender

Discount User

Spend Level

Above Average

Engagement

Non-Subscriber

Loyalty Segments



New Customers

Initial purchasers



Returning Customers

Repeat purchasers



Loyal Customers

Frequent purchasers

Core Spending Insights

Revenue Drivers · Demographic Value · Behavioral Patterns



Gender Revenue

Male vs Female

2x

Dominant Segment

Male



Subscription Spend

Higher Spenders

Non-Sub

Revenue Potential

Untapped



Age Group Revenue

Top Generator

Young

Revenue Spread

Even

Product & Behavior Insights

Key Findings from 3,900 Transactional Records



Top Products

Jewelry & Blouse lead with **171 orders** each



Most Reviewed

Gloves are the **most reviewed** product in store



Discount Users

839 users spent above average despite using discounts



Shipping Preference

Express shipping leads in **average amount**



Customer Loyalty

Segmented into **New, Returning, Loyal** groups



Repeat Buyer Status

1/3 of repeat buyers (>5 purchases) are subscribed

Insight Focus: High-value discount users and product category performance drive strategic placement decisions

Analysis Methodology

Data Preparation • Feature Engineering • SQL Analysis • Visualization

01

Preparation



Data Prep

-  Standardize column names
-  Handle missing values (Median)
-  Remove redundant features

02

Engineering



Feature Eng.

-  Add age_group segments
-  Numerical purchase frequency
-  Snake_case formatting

03

Analysis



SQL Insights

-  Gender revenue split
-  Customer loyalty tiers
-  Top product categories

04

Visualization



Dashboard

-  PowerBI integration
-  Interactive visual summaries
-  Trend analysis at a glance

 **Data Integrity:** Rigorous cleaning and transformation ensure reliable insights from 3,900 transaction records



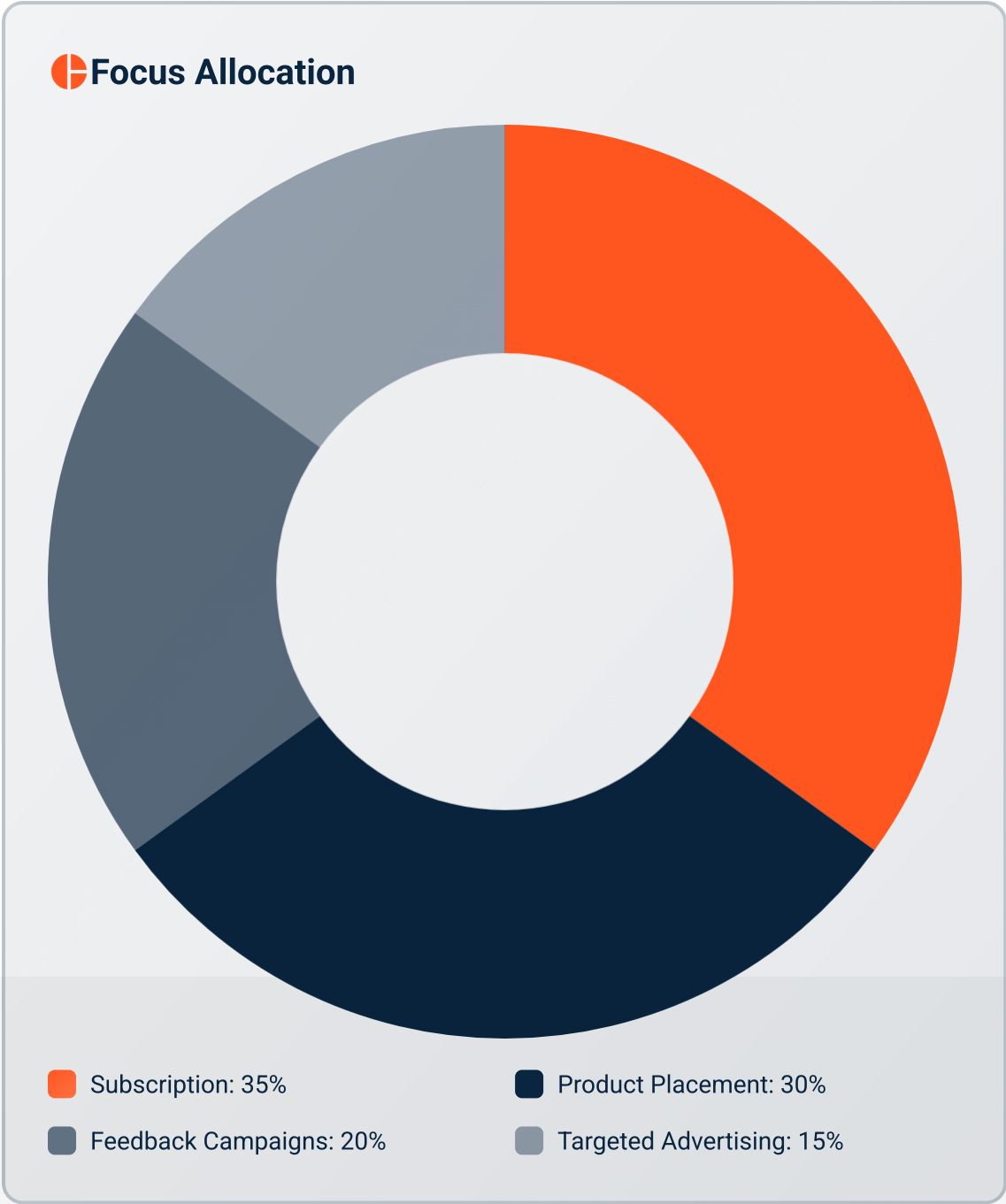
Demographic & Spending Drivers

Key Revenue Influencers Across Customer Segments

 Gender	 Age Group	 Subscription	 Shipping
Male Revenue 2x Higher	Top Revenue Young Adults	Highest Spenders Non-Subscribed	Preference Express Leads
Female Segment Growth Opportunity	Revenue Gap Not Substantial	Repeat Buyers 1/3 Subscribed	Margin Small Difference
Insight Targeted Ads Needed	Strategy Maintain All Ages	Action Conversion Campaign	Opportunity Premium Services
 Male Revenue Driver 2x higher spending capacity	 Non-Subscriber Value Higher average and total spend	 Repeat Buyer Upsell Convert high-frequency shoppers	

Strategic Recommendations

Focus Areas for Revenue Growth & Customer Retention



Actionable Insights



Subscription Campaigns

High Priority

Target non-subscriber big spenders with conversion offers

✓ Action: Exclusive discounts & loyalty perks



Product Placement

Strategic

Prioritize clothing & accessories display for high-revenue items

✓ Action: Feature Jewelry & Blouses prominently



Feedback Surveys

Quality

Launch survey campaigns to improve average rating (3.75/5)

✓ Action: Incentivize reviews for under-reviewed items



Female-Centric Ads

Growth

Implement ad campaigns targeting female shoppers with incentives

✓ Action: Personalized promotions & category targeting



The following slide highlights strategic recommendations for achieving long-term business objectives. It includes various steps such as define strategic position, evaluate business goals, build strategic plan, etc

Define strategic position

- » Determine issues to be solved
- » Conduct SWOT analysis to prepare better business strategies
- » Conduct PEST analysis for understanding current scenario
- » Text Here

Build strategic plan

- » Determine strategies to achieve long-term organizational goals
- » Implement strategic decisions to achieve strategic position and desired value
- » Define responsibilities clearly to staff
- » Text Here

Plan review

- » Perform audit to review performance
- » Reevaluate business objectives to make changes in existing plans
- » Use KPI dashboards to track performance
- » Text Here

Driving Growth

Actionable Insights for Sustainable Revenue & Customer Value

Male Revenue Dominance

2x Higher

Non-Subscriber Value

Top Spenders

Top Products

Jewelry & Blouse

Evaluate business goals

- » Define mission and vision
- » Determine company's competitive edge
- » Formulate organization-wide strategy to achieve business goals
- » Text Here



Contact Us
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Plan management and implementation

- » Communicate strategies to internal management and employees
- » Provide training to staff to implement
- » Convert whole strategy into concrete plan
- » Text Here



Next Steps
Q3 Strategy Review